



Adult nursing

Oncology nursing management in cancer care

المحاضرة النظرية -

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Oncology is a branch of medicine that deals with the prevention, diagnosis, and treatment of cancer.

Benign: A tumor that is not cancerous. The tumor does not invade nearby tissue or spread to other parts of the body

Carcinoma: Cancer that starts in skin or tissues that line the inside or cover the outside of internal organs.

Chemotherapy: Drugs used to destroy cancer cells by interfering with their growth and/or preventing their reproduction.

Malignant: A tumor that is cancerous. The tumor may invade nearby healthy tissue or spread to other parts of the body.

Metastasis: The spread of cancer from where the cancer began to another part of the body. Cancer cells can break away from the primary tumor and travel through the blood or the lymphatic system to the lymph nodes, brain, lungs, bones, liver, or other organs.

Palliative: Treatment of the physical, spiritual, psychological, and social needs of a person with cancer. Its purpose is to improve quality of life.

Adjuvant therapy: Treatment given after the main treatment. It usually refers to chemotherapy, radiation therapy, hormone therapy, or immunotherapy given after surgery to reduce the

chance of cancer coming back.

Hormone therapy: Treatment that removes, blocks, or adds hormones to kill or slow the growth of cancer cells. (Also called hormonal therapy or endocrine therapy.)

Thrombocytopenia نقص الصفيحات — Low platelet count

Radiation therapy: The use of high-energy rays (such as x-rays) to kill or shrink cancer cells.

Recurrence: Cancer that has returned after a period of time when the cancer could not be detected.

Remission: The disappearance of the signs and symptoms of cancer but not necessarily the entire disease

Neoplasia: refers to a mass that has developed due to abnormal cell or tissue growth.

Introduction

2nd most common cause of death

5-year survival rate is now 62% for those who are disease free, in remission تخفيف , or under treatment

In 2019, nearly 8 million people will die of cancer, and left unchecked, the number of deaths will increase to 13.2 million per year by 2030.

Incidence of cancer

- Cancer affect people of all ages, but most cases (76%) are diagnosed in those over the age of (55) years.
- Cancer incidence is higher in blacks than in whites and other minority groups.
- Lung cancer is leading cause of cancer-related death in both men & women.
- Other cancers, such as breast cancer, prostate, occur more often than lung cancer.
- American Cancer Society (ACS) indicates that in the United States men have 1-2 life time risk of developing cancer, for women the risk 1-3.
- The leading cancer sites for the male are prostate, lung, colon, and rectum; the leading cancer sites for women are breast, lung, colon, and rectum.
- Every five deaths in the United States one is from cancer. Making it second leading cause of death (heart disease is the most common).

What Is Cancer?

Cancer can start any place in the body. It starts when cells grow out of control and crowd out normal cells. This makes it hard for the body to work the way it should.

Cancer basics

Cancer is not just one disease.

There are many types of cancer. Cancer can start in the lungs, the breast, the colon, or even in the blood. Cancers are alike in some ways, but they are different in the ways they grow and spread.

Many cancers form solid tumors, which are masses of tissue. Cancers of the blood, such as leukemias, generally do not form solid tumors.

How are cancers alike?

The cells in our bodies all have certain jobs to do. Normal cells divide in an orderly way. They die when they are worn out or damaged, and new cells take their place. Cancer is when the cells start to grow out of control. The cancer cells keep on growing and making new cells. They crowd out normal cells. This causes problems in the part of the body where the cancer started.

When Cancer Spreads

Cancer cells can also spread to other parts of the body through the blood or the lymph system and form new tumors far from the original tumor. For example, cancer cells in the lung can travel to the bones and grow there. When cancer cells spread, it's called metastasis (meh-TAS-tuh-sis). When lung cancer spreads to the bones, it's still called lung cancer, the cancer cells in the bones look just like the ones from the lung. It's not called bone cancer unless it started in the bones..

How Cancer Arises

Cancer is a genetic disease—that is, it is caused by changes to genes that control the way our cells function, especially how they grow and divide.

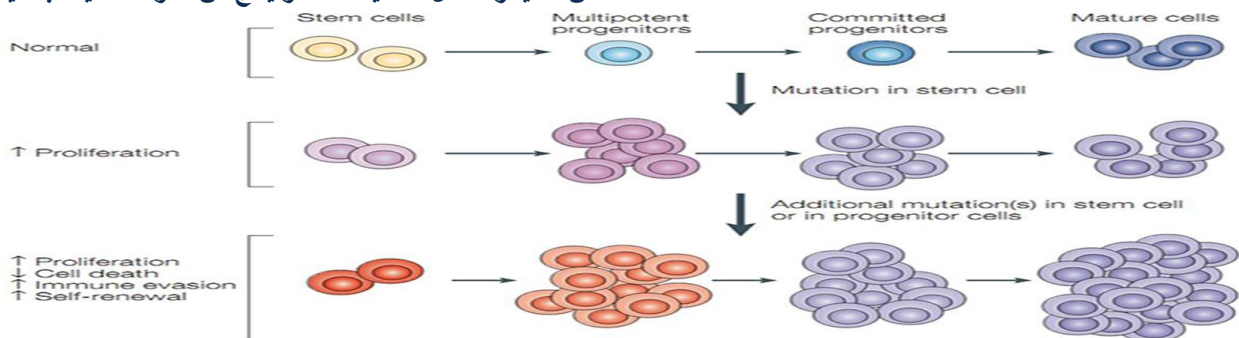
Genetic changes that cause cancer can be inherited from our parents. They can also arise during a person's lifetime as a result of errors that occur as cells divide or because of damage to DNA caused by certain environmental exposures. Cancer-causing environmental exposures include substances, such as the chemicals in tobacco smoke, and radiation, such as ultraviolet rays from the sun

Defect in Cellular Proliferation

Stem cell theory نظرية الخلايا الجذعية

Loss of intracellular control of proliferation results from mutation of stem cells

فقدان السيطرة داخل الخلايا للانتشار ينتج من طفرة الخلايا الجذعية



How are cancers different?

Some cancers grow and spread fast. Others grow more slowly. They also respond to treatment in different ways. Some types of cancer are best treated with surgery; others respond better to drugs called chemotherapy (key-mo-THER-uh-pee). Often 2 or more treatments are used to get the best results.

What are tumors?

Most cancers form a lump called a tumor or a growth. But not all lumps are cancer. Doctors take out a piece of the lump and look at it to find out if it's cancer. Lumps that are not cancer are called benign (be-NINE). Lumps that are cancer are called malignant (muh-LIG-nunt).

characteristics of benign and malignant neoplasms

Characteristics of Benign and Malignant Neoplasms		
Characteristics	Benign	Malignant
Metastasis	Negative	Gains access to the blood and lymphatic channels and metastasizes to other areas of the body
General effects	Is usually a localized phenomenon that does not cause generalized effects unless its location interferes with vital functions	Often causes generalized effects, such as anemia, weakness, and weight loss
Tumor destruction	Does not usually cause tissue damage unless its location interferes with blood flow	Often causes tissue damage as the tumor outgrows its blood supply or encroaches on blood flow to the area; may also produce substances that cause cell damage
Ability to cause death	Does not usually cause death unless its location interferes with vital functions	Usually causes death unless growth can be controlled

Characteristics of Benign and Malignant Neoplasms		
Characteristics	Benign	Malignant
Cell characteristics	Well-differentiated that resemble normal cells of the tissue from which the tumor originated	Cells are undifferentiated and often bear little resemblance to the normal cells of the tissue from which they arose
Mode of growth	Tumor grows by expansion and does not infiltrate the surrounding tissues; usually encapsulated	Grows at the periphery and sends out processes that infiltrate and destroy the surrounding tissues
Rate of growth	Usually slow	Variable and depends on level of differentiation; the more anaplastic the tumor, the faster its growth

Carcinogenic Agents and Factors (ETIOLOGY)

- Viruses and bacteria
- Physical factors: sunlight, radiation, chronic irritation
- Chemical agents: tobacco, asbestos
- Genetic and familial factors
- Diet Hormones
- Role of the immune system

Development of Cancer

Malignant transformation needs 3 steps

• 1- Initiation

- Mutation of genetic structure طفرة في البنية الجينية
- Has potential to develop into clone of neoplastic cells لديه القدرة على التطور إلى استنساخ الخلايا الورمية

2- Promotion

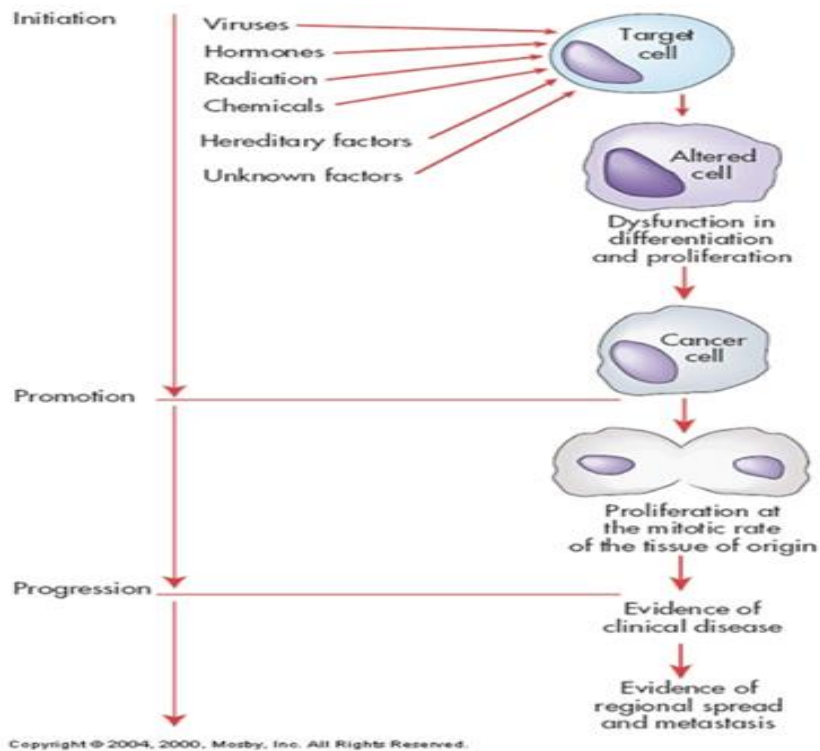
- Characterized by the reversible proliferation of altered cells تتميز بتكاثر الخلايا القابلة للتغيير
- Activities of promotion (e.g. obesity, smoking, alcohol) are reversible أنشطة الترويج (مثل السمنة ، التدخين ، الكحول) يمكن عكسها
- Latent period فترة الكامنة

Initial genetic alteration to clinical evidence of cancer

3- Progression

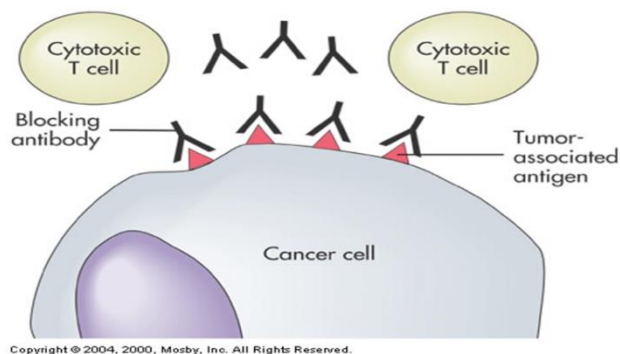
- Characterized by increased growth rate of tumor as well as its invasiveness and metastasis غزو
- Metastasis = spread of cancer from primary (initial) site to distant site
- Metastasis process begins with rapid growth of primary tumor
- Tumor angiogenesis - formation of blood vessels within the tumor; critical for tumor survival
- الأورام والأوعية الدموية - تشكيل الأوعية الدموية داخل الورم. خطرة لبقاء الورم

Process of Cancer Development



Role of Immune System

- Cancer is related to failure of normal immune system
- Cancer cells has surface antigen → T lymphocytes (the soldier of the immune response) → proliferate and kill cancer cells (cytotoxic effect)
- Tumore cells secrete a substance that destroy body defences → increase susceptibility to infection in cancer patients



Primary and Secondary Prevention

Primary prevention is concerned with reducing cancer risk in healthy people

- Secondary prevention involves detection and screening to achieve early diagnosis and intervention.
- There is now great emphasis on the primary and secondary prevention of cancer.

Primary Prevention Avoid known carcinogens.

- Lifestyle and dietary changes to reduce cancer risk
- Public and patient education

Secondary Prevention Identification of patients at high cancer risk

- Cancer screening
- Self-breast exam
- Self-testicular
- Exam Screening
- colonoscopy
- Pap test
- Public and patient education

Classification of Cancer

A-Anatomic Site Classification تصنيف الموقع التشريحي

Identified by

- tissue origin أصل الأنسجة
- anatomic site موقع تشريحي

behavior of the tumor (benign vs. malignant)

Anatomic Site Classification

- **Carcinomas** originate from embryonal ectoderm(skin ,gland) and endoderm(mucous membrane lining respiratory tract, GI)
- **Sarcomas** originate from embryonic mesoderm(muscle, bone ,fat ,connective tissue)
- **Lymphomas and leukemias** originate from hematopoietic system
- **Multiple Myeloma** is cancer that begins in [plasma cells](#), another type of immune cell

Classification (TNM)

- Tumor size
- Spread to lymph nodes
- Metastasis

Clinical Staging

- 0: cancer in situ

- 1: tumor limited to tissue of origin
- 2: limited local spread
- 3: extensive local and regional spread
- 4: metastasis

TNM Cancer Staging Classification System

Classification	Staging	Tissue Involvement
1. Primary tumor (T)		
Tis	Stage I	Tumor in situ, indicates no invasion of other tissues
T1, T2, T3, T4	Stage II	Ranges indicate progressive increase in tumor size with local metastasis
2. Regional lymph node involvement (N)		
N0	No Nodes	
N1, N2, N3, N4	Stage III	Metastasis to regional lymph nodes
3. Metastasis (M)		
M0		No metastasis
M1	Stage IV	Distant metastasis

Early Warning Signs for Cancer

Cancer 7 early warning signs (CAUTION)

1. Change in bowel or bladder habits.
2. A sore that does not heal.
3. Unusual bleeding or discharge.
4. Thickening or lump in breast or scrotum.
5. Indigestion or difficulty in swallowing
6. Obvious change in a mole or wart.
7. Nagging cough or hoarseness.

Diagnosis of Cancer Diagnostic Tests

Biopsy involves histologic examination by a pathologist of a piece of tissue

- **incisional biopsy** – part of the tumour is cut out
- **excisional biopsy** – the entire tumour is cut out. A margin of healthy tissue is usually removed at the same time
- **core biopsy** – a needle is used to remove a thin core of tissue. Sometimes a CT scan or ultrasound is used to guide the needle

- **fine needle aspiration or biopsy** – a thin needle is used to remove a sample of fluid and tissue from the tumour

Imaging Tests Used to Detect Cancer

TEST	DESCRIPTION	DIAGNOSTIC USE
Tumor markers and cancer	Analysis of substances found in blood or other body fluids that are made by the tumor or by the body in response to the tumor	Breast, colon, lung, ovarian, testicular, prostate cancers
Magnetic resonance imaging(MRI)	Use of magnetic fields and radiofrequency signals to create sectioned images of various body structures	Neurologic, pelvic, abdominal, thoracic cancers
Computed tomography (CT scan)	Use of narrow beam x-ray to scan successive layers of tissue for a cross-sectional view	Neurologic, pelvic, skeletal, abdominal, thoracic cancers
Fluoroscopy	Use of x-rays that identify contrasts in body tissue densities; may involve the use of contrast agents	Skeletal, lung, gastrointestinal cancers
Ultrasonography (ultrasound)	High-frequency sound waves echoing off body tissues are converted electronically into images; used to assess tissues deep within the body	Abdominal and pelvic cancers
Endoscopy	Direct visualization of a body cavity or passageway by insertion of an endoscope into a body cavity or opening; allows tissue biopsy, fluid aspiration and excision of small tumors; both diagnostic and therapeutic	Bronchial, gastrointestinal cancers

Nuclear medicine imaging	Uses intravenous injection or ingestion of radioisotope substances followed by imaging of tissues that have concentrated the radioisotopes	Bone, liver, kidney, spleen, brain, thyroid cancers
Positron emission tomography (PET scan)	Computed cross-sectional images of increased concentration of radioisotopes in malignant cells provide information about biologic activity of malignant cells; help distinguish between benign and malignant processes and responses to treatment	Lung, colon, liver, pancreatic, breast, esophagus cancers; Hodgkin's and non-Hodgkin's lymphoma and melanoma

Barium Enema.

Bone Marrow Aspiration and Biopsy.

Bone Scan

Colonoscopy

Digital Rectal Exam (DRE)

Fecal Occult Blood Tests

Mammography

Pap Test

Collaborative Care of Cancer

Goals:

- Cure
- Control
- Palliation

Common Treatments for Cancer

- Surgery
- Radiation
- Chemotherapy
- Biologic Therapy
- Bone marrow or stem cell transplant

Surgical therapy

- to cure or control
- Slow cancers are most amenable
- Margin of normal tissue must surround tumor

Types of surgeries include:

Curative surgery. removes the cancerous tumor or growth from the body. ...

Diagnostic surgery. ...

Debulking surgery. removes a portion, though not all, of a cancerous tumor...

Palliative surgery. is used to treat cancer at advanced stages...

Supportive surgery. ...

Restorative surgery. Restorative surgery is sometimes used as a follow-up to curative or other surgeries to change or restore a person's appearance or the function of a body part. جراحة اصلاحية.

Nursing management in cancer surgery

Pre op assessment for factors that affect pt. Education and emotional support Communicate frequently with Dr and other health care member to be certain that provided information is consistent After surgery the nurse assess pt response to surgery, and monitor him for complication (infection, bleeding, thrombophlebitis, fluid and electrolyte imbalance, Teach pt about wound care, activity, nutrition and medications)

Radiation therapy

- Emission and distribution of energy through space or material medium
- Energy produced breaks bonds in DNA, leading to death at time of reproduction
- Affects both cancer as well as normal cells
- Normal tissues are usually able to recover

Radiation Therapy: Curative, control, or palliative

Types of Radiation Therapy:

External radiation

Internal radiation

To decrease exposure to radiation Pt placed in private room, the staff member wear protective clothing, hr's of visit limits to 30min/day , and 6 feet distance

Nursing Care of the Patient Undergoing Radiation Therapy

Patient and family education Include restrictions and precautions Skin care Oral care Protection of care providers

Toxicity from radiation therapy

The effect: Altered skin integrity (alopecia/fall of hair, erythema) Altered oral mucosa (stomatitis, xerostomia /mouth dryness, change & taste loss, decrease salivation) Esophageal irritation with chest pain and dysphasia ,Anorexia, nausea and vomiting & diarrhea ,Anemia, leukopenia and thrombocytopenia ,Fatigue and malaise

Nursing Management: Patients Undergoing Radiation

- **Fatigue**
 - Possibly due to accumulation of metabolites from cell destruction
 - Rest before activity
 - Get assistance with activity
 - Maintain nutritional status
- **Anorexia**
 - Monitor carefully to avoid weight loss
 - Weigh twice weekly
 - Small, frequent, high-protein, high-calorie meals
 - Supplements
- **Skin Reactions**
 - Occurs within treatment field
 - Lubricate dry skin with nonirritating lotion or solution (no metal, alcohol, perfume, or additives)
 - Wet desquamation must be kept clean and protected from further damage
 - Prevention of infection
 - Facilitate wound healing
 - Protect irritated skin from extremes in temperature
 - Avoid constricting garments, harsh chemicals, and deodorants
- **Bone Marrow Suppression**
 - If bone marrow is within treatment field
 - Kills RBCs, WBCs, platelets
 - Not as significant as with chemotherapy
 - Monitor blood counts
 - May need transfusion for anemia
- **Oral, Oropharynx, and Esophageal Reactions**
 - Teach patients to examine oral cavity
 - Dental work before initiation of radiation therapy
 - Saliva substitutes for dry mouth
 - Oral care (brushing and flossing unless contra-indicated)
 - Pain relief

- Frequent feedings of soft, nonirritating, high-protein, high-calorie foods
 - Avoidance of extremes in temperature, alcohol, and tobacco
- **Pulmonary Effects** – pneumonitis (cough, SOB, fever, night sweats)
 - Treatment
 - Bronchodilators
 - Expectorants/cough suppressants
 - Bed rest
 - Oxygen
- **Gastrointestinal Effects** – ↓ secretion of HCl, mucus, pepsin → N&V, diarrhea
 - Prophylactic administration of antiemetics
 - Assess for S/S of alkalosis and dehydration
 - I&O
 - nonirritating diet
 - Antidiarrheal medications
- **Reproductive Effects**
 - Risk of infertility
 - Inform patient on expected sexual side effects
 - Consider harvesting sperm or ova
 - Refer to counseling if needed
- **Coping**
 - Assist in planning for transportation, nutrition, and emotional support
 - Patient teaching of symptom management to maintain highest possible quality of life

Chemotherapy

Chemotherapy Agents used to destroy tumor cells by interfering with cellular function and replication Curative, control, or palliative Used to treat systemic disease rather than localized lesion.

Dosage depends on:

Patients total body surface area

Physical status

Previous response to chemotherapy

Classification of Chemotherapeutic Agents

- cell cycle–specific drugs: destroy cells actively reproducing by means of the cell cycle. Many of these agents are specific to certain phases of the cell cycle. Most affect cells in the S phase by interfering with DNA and RNA synthesis. Others, such as the vinca or plant alkaloids, are specific to the M phase, where they halt mitotic spindle formation.
- cell cycle–nonspecific agents: These agents usually have a prolonged effect on cells, leading to cellular damage or death.

Methods of Administration

Oral

IM

IV

Intrathecal – into subarachnoid space via lumbar puncture

Intraarterial – into artery the supplies tumor

Intravesical (bladder)

Intraperitoneal

Antineoplastic Agents

DRUG CLASS AND EXAMPLES	MECHANISM OF ACTION	COMMON SIDE EFFECTS
Alkylating Agents melphalan, nitrogen mustard, thiotepa	Alter DNA structure by misreading DNA code, initiating breaks in the DNA molecule, cross-linking DNA strands	Bone marrow suppression, nausea, vomiting, cystitis (cyclophosphamide, ifosfamide), stomatitis, alopecia, gonadal suppression, renal toxicity (cisplatin)
Nitrosureas carmustine (BCNU) , streptozocin	Similar to the alkylating agents; cross the blood–brain barrier	Delayed and cumulative myelosuppression, especially thrombocytopenia; nausea, vomiting
Topoisomerase I Inhibitors irinotecan, topotecan	Induce breaks in the DNA strand by binding to enzyme topoisomerase I, preventing cells from dividing	Bone marrow suppression, diarrhea, nausea, vomiting, hepatotoxicity
Antimetabolites 5-azacytadine, cytarabine, 5-fluorouracil (5-FU)	Interfere with the biosynthesis of metabolites or nucleic acids necessary for RNA and DNA synthesis	Nausea, vomiting, diarrhea, bone marrow suppression, proctitis, stomatitis, renal toxicity (methotrexate), hepatotoxicity
Antitumor Antibiotics bleomycin, dactinomycin,	Interfere with DNA synthesis by binding DNA; prevent RNA synthesis	Bone marrow suppression, nausea, vomiting, alopecia, anorexia, cardiac toxicity

Mitotic Spindle Poisons Plant alkaloids: etoposide	Arrest metaphase by inhibiting mitotic tubular formation (spindle); inhibit DNA and protein synthesis	Bone marrow suppression (mild with VCR), neuropathies (VCR), stomatitis
Hormonal Agents androgens and antiandrogens estrogens and antiestrogens,	Bind to hormone receptor sites that alter cellular growth; block binding of estrogens to receptor sites (antiestrogens); inhibit RNA synthesis; suppress aromatase of P450 system, which decreases estrogen level	Hypercalcemia, jaundice, increased appetite, masculinization, feminization, sodium and fluid retention, nausea, vomiting, hot flashes, vaginal dryness
Miscellaneous Agents asparaginase, procarbazine		Anorexia, nausea, vomiting, bone marrow suppression, hepatotoxicity, anaphylaxi

Chemotherapy :Effects on Normal Tissues

- Chemotherapeutic agents cannot distinguish between normal and cancer cells
- Body's response to products of cellular destruction in circulation may cause fatigue, anorexia, and taste alterations
- **Acute toxicity**
 - Vomiting
 - Allergic reactions
 - Arrhythmias
- **Delayed effects**
 - Mucositis
 - Alopecia
 - Bone marrow suppression
- **Chronic toxicities**
 - Damage to
 - Heart
 - Kidney
 - Liver
 - Lungs

Gastrointestinal System: Nausea and vomiting are the most common side effects of chemotherapy and may persist for up to 24 hours after its administration.

Hematopoietic System. Most chemotherapeutic agents cause myelosuppression (depression of bone marrow function), resulting in decreased production of blood cells. Myelosuppression decreases the number of WBCs (leukopenia), red blood cells (anemia), and platelets (thrombocytopenia) and increases the risk for infection and bleeding.

Reproductive system: Testicular and ovarian function can be affected by chemotherapeutic agents, resulting in possible sterility

Renal System. damage the kidneys because of their direct effects during excretion and the accumulation of end products after cell lysis(uric acid)
Monitoring blood urea nitrogen, serum creatinine, creatinine clearance, and serum electrolyte levels is essential. Adequate hydration, alkalinization of the urine to prevent formation of uric acid crystals.

Cardiopulmonary System. Antitumor antibiotics can cause cumulative cardiac toxicities, cumulative toxic effects on lung function can cause Pulmonary fibrosis .
Neurologic System. Peripheral neuropathies, loss of deep tendon reflexes, and paralytic ileus may occur.

Nursing Management in Chemotherapy

The nurse has an important role in assessing and managing many of the problems.

- Assess pt. nutritional and fluid and electrolyte status due to Anorexia, nausea, vomiting, altered taste, and diarrhea.
- Use creative way to improve appetite.
- Due to depression of bone marrow and immune system pt. will be liable to anaemia , infection , bleeding disorders,
- Identify factors that further increase this risk asepsis and gentle handling.
- Monitor blood cell count closely
- Report untoward blood test results ,signs of infection and bleeding
- Instruct family members about previous measures at home.
- Observe for risk and consequences of extravasation and report it immediately if it occurred.
- Handling of chemotherapy agent: exposure of nurse to small dose by contact , inhalation, ingestion may effect .persecution and preparation

Nursing care of the patient with cancer

Nursing process:

Assessment

- Assess factors predisposing to bleeding or /and infection

- Monitor lab results esp. CBP
- Inspect common sites for infection and bleeding, pharynx, skin, urinary, respiratory.
- Assess for sepsis especially with invasive procedures
- Monitor for bleeding
 - skin and tissue integrity may be alerted in ca due to :
 - Skin lesion and ulceration
 - Oral mucous membrane lesion
 - Alopecia
- Nutritional state alteration and weight loss (nausea, vomiting ,diarrhea)-caloric, protein serum
 - Pain and discomfort
 - Psychological and mental state

Nursing Diagnosis:

- **Risk for infection related to altered immunologic response**
- **Impaired skin integrity: erythematous and wet desquamation reactions to radiation therapy .**
- **Impaired oral mucous membrane: stomatitis**
Impaired tissue integrity: alopecia.
- **Imbalanced nutrition, less than body requirements, related to nausea and vomiting.**
- **Imbalanced nutrition: less than body requirements, related to anorexia, cachexia, or malabsorption.**
- **Fatigue**
- **Chronic Pain.**
- **grieving related to anticipated loss**
- **Disturbed body image and situational low self-esteem related to changes in appearance, function, and roles.**

Potential complication: risk for bleeding problems

Nursing goals and Interventions

- **Nursing Diagnosis: Risk for infection related to altered immunologic response**

Goal: Prevention of infection

1. Assess patient for evidence of infection:

a. Check vital signs every 4 hours.b. Monitor WBC count and differential each day.c.

Inspect all sites that may serve as entry ports for pathogens (intravenous sites, wounds, skin folds, bony prominences, perineum, and oral cavity).

2. Report fever $\geq 38.3^{\circ}\text{C}$ (101°F), chills, diaphoresis, swelling, heat, pain, erythema, exudate on any body surfaces. Also report change in respiratory or mental status, urinary

frequency or burning, malaise, myalgias, arthralgias, rash, or diarrhea.

3. Obtain cultures and sensitivities as indicated before initiation of antimicrobial treatment (wound exudate, sputum, urine, stool, blood)

4. Initiate measures to minimize infection.

Discuss with patient and family: Placing patient in private room. Avoiding contact & use PPE. Avoid rectal or vaginal procedures(rectal temperatures, examinations, suppositories; vaginal tampons). Use stool softeners to prevent constipation and straining. Instruct patient to use electric razor. Avoid fresh fruits, raw meat, fish, and vegetables.

5. Assess intravenous sites .

6. Avoid intramuscular injections.

7. Avoid insertion of urinary catheters

Collaborative Problem: Potential complication: risk for bleeding problems

Goal: Prevention of bleeding

1. Assess for potential for bleeding: monitor platelet count.

2. Assess for bleeding: Petechiae or ecchymosis, Decrease in Hb .Prolonged bleeding from invasive procedures, venepunctures, minor cuts or scratches, Frank or occult blood in any body excretion, emesis, sputum. Bleeding from any body orifice

3. Instruct patient and family about ways to minimize bleeding:

Use soft toothbrush for mouth care. Avoid commercial mouthwashes.

Use electric razor for shaving. Use emery board for nail care. Avoid foods that are difficult to chew.

4. Initiate measures to minimize bleeding: Draw all blood for lab work with one daily venepuncture. Avoid taking temperature rectally or administering suppositories and enemas . Avoid intramuscular injections; use smallest needle possible. Apply direct pressure to injection and venepuncture sites for at least 5 min. Lubricate lips with petrolatum.

Avoid bladder catheterizations; use smallest catheter if catheterization is necessary. Maintain fluid intake of at least 3 L/24 h unless contraindicated. Use stool softeners or increase bulk in diet. Avoid medications that will interfere with clotting (eg, aspirin).

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Nursing Diagnosis: Impaired tissue integrity: alopecia

Goal: Maintenance of tissue integrity; coping with hair loss

1. Discuss potential hair loss and regrowth with patient and family.
2. Explore potential impact of hair loss on self-image, interpersonal relationships, and sexuality.
3. Prevent or minimize hair loss through the following:
 - a. Use scalp hypothermia and scalp tourniquets, if appropriate.
 - b. Cut long hair before treatment.
 - c. Use mild shampoo and conditioner, gently pat dry, and avoid excessive shampooing.
 - d. Avoid electric curlers, curling irons, dryers, clips, barrettes, hair sprays, hair dyes, and permanent waves.
 - e. Avoid excessive combing or brushing; use wide-toothed comb.
4. Prevent trauma to scalp.
 - a. Lubricate scalp with vitamin A&D ointment to decrease itching.
 - b. Have patient use sunscreen or wear hat when in the sun.
5. Suggest ways to assist in coping with hair loss:
 - a. Purchase wig or hairpiece before hair loss.
 - b. Begin to wear wig before hair loss.
 - c. Wear hat, scarf, or turban.
6. Encourage patient to wear own clothes and retain social contacts.
7. Explain that hair growth usually begins again once therapy is completed.

Nursing Diagnosis: Impaired skin integrity: erythematous and wet desquamation reactions to radiation therapy

Goal: Maintenance of skin integrity

1. In erythematous areas:
 - a. Avoid the use of soaps, cosmetics, perfumes, powders, lotions and ointments, deodorants.
 - b. Use only lukewarm water to bathe the area.
 - c. Avoid rubbing or scratching the area.
 - d. Avoid shaving the area with a straight edged razor.
 - e. Avoid applying hot-water bottles, heating pads, ice, and adhesive tape to the area.

- f. Avoid exposing the area to sunlight or cold weather.
- g. Avoid tight clothing in the area. Use cotton clothing.
- h. Apply vitamin A&D ointment to the area.
2. If wet desquamation occurs:
 - a. Do not disrupt any blisters that have formed.
 - b. Avoid frequent washing of the area.
 - c. Report any blistering.
 - d. Use prescribed creams or ointments.
 - e. If area weeps, apply a thin layer of gauze dressing

Nursing Diagnosis: Imbalanced nutrition: less than body requirements, related to anorexia, cachexia, or malabsorption

Goal: Maintenance of nutritional status

1. Teach patient to avoid unpleasant sights, odors, sounds in the environment during mealtime.
2. Suggest foods that are preferred preferably high-calorie and high-protein
3. Encourage adequate fluid intake, but limit fluids at mealtime.
4. Suggest smaller, more frequent meals.
5. Promote relaxed, quiet environment during mealtime
6. Consider cold foods, if desired.
7. Advocate nutritional supplements and high-protein foods between meals.
8. Encourage frequent oral hygiene.
9. Provide pain relief measures.
10. Provide control of nausea and vomiting.
11. Increase activity level as tolerated.
12. Decrease anxiety by encouraging verbalization of fears, concerns; use of relaxation techniques; imagery at mealtime.
13. Position patient properly at mealtime.
14. For collaborative management, provide enteral tube feedings of commercial liquid diets, elemental diets.
15. Provide parenteral nutrition

Nursing Diagnosis: Chronic Pain

Goal: Relief of pain and discomfort

1. Use pain scale to assess pain and discomfort characteristics: location, quality, frequency, duration, etc
2. Assure patient that you know that pain is real and will assist him or her in reducing it.
3. Assess other factors contributing to patient's pain: fear, fatigue, anger, etc.
4. Administer analgesics to promote optimum pain relief within limits of physician's prescription.
5. Assess patient's behavioral responses to pain and pain experience.
6. Collaborate with patient, physician, and other health care team members when

changes in pain management are necessary.

7. Encourage strategies of pain relief that patient has used successfully in previous pain experience.

8. Teach patient new strategies to relieve pain and discomfort: distraction, imagery, relaxation, cutaneous stimulation, etc

Late Effects of Radiation and Chemotherapy

Cancer survivors at risk for leukemia and other secondary malignancies

Biologic Therapy

Biological therapy involves the use of living organisms, substances derived from living organisms, or laboratory-produced versions of such substances to treat disease. Some biological therapies for cancer stimulate the body's immune system to act against cancer cells. These types of biological therapy, which are sometimes referred to collectively as "immunotherapy," do not target cancer cells directly. Other biological therapies, such as antibodies, do target cancer cells directly.

Alters biologic response to tumor cells

- Direct anti-tumor effect
- Restore, augment, or modulate immune system
- Other effects – interfere with metastasis, differentiation

Bone Marrow and Stem Cell Transplantation

- Allows for safe use of very high doses of chemotherapy or radiation therapy
- Procedure with many risks, including death
- Highly toxic
- Allogeneic
 - From donor
 - Goal is to administer large doses of systemic therapy
 - Then "rescue" bone marrow through engraftment and subsequent normal proliferation and differentiation of donated marrow
- Autologous
 - Patient receives their own bone marrow
 - Marrow is removed, treated, stored, and reinfused
- Syngeneic
 - Obtaining stem cells from one identical twin and infusing them into the other
- Harvesting

Procedure conducted in the OR
Multiple aspirations carried out
Usually iliac crest or sternum
May be treated to remove cancer cells if autologous
Cryopreserved

- Complications
 - Bacterial, viral and fungal infections are common
 - Prophylactic antibiotic therapy
 - Graft-versus-host disease
 - Lymphocytes from donated marrow recognize recipient as foreign
 - Attack organs such as skin, liver, and intestines

Management of Cancer Pain

- Patient report should always be believed and accepted as primary pain assessment data
- Drug therapy should be used following WHO analgesic ladder
- Nonpharmacologic interventions can be effectively used

Cancer: Psychologic Support

- **Emphasis placed on maintaining optimal quality of life**
- **Positive attitude of patient, family, and health care providers has significant positive impact on quality of life for patient**
- **May also influence prognosis**
- **Continue to be available**
- **Exhibit caring attitude**
- **Listen actively to fears and concerns**
- **Provide relief from distressing symptoms**
- **Maintain relationship based on trust and confidence**
- **Use touch to exhibit caring**
- **Assist patient in setting realistic short- term goals**
- **Assist in maintaining usual lifestyle patterns**
- **Maintain hope, which can vary**
- **Provides control over what is occurring**
- **Basis of positive attitude**

Prevention

There's no certain way to prevent cancer. But doctors have identified several ways of reducing your cancer risk, such as:

- **Stop smoking.** If you smoke, quit. If you don't smoke, don't start. Smoking is linked to several types of cancer — not just lung cancer. Stopping now will reduce your risk of cancer in the future.

- **Avoid excessive sun exposure.** Harmful ultraviolet (UV) rays from the sun can increase your risk of skin cancer. Limit your sun exposure by staying in the shade, wearing protective clothing or applying sunscreen.
- **Eat a healthy diet.** Choose a diet rich in fruits and vegetables. Select whole grains and lean proteins.
- **Exercise most days of the week.** Regular exercise is linked to a lower risk of cancer. Aim for at least 30 minutes of exercise most days of the week. If you haven't been exercising regularly, start out slowly and work your way up to 30 minutes or longer.
- **Maintain a healthy weight.** Being overweight or obese may increase your risk of cancer. Work to achieve and maintain a healthy weight through a combination of a healthy diet and regular exercise.